WORKED SOLUTION

STEP III 1994, Question 2

Pure • Palindromic Quartic Equations

STEP III 1994, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q2

The key idea

The quartic in (i) is *palindromic*: its coefficients read the same forwards and backwards. Dividing by x^2 pairs x^2 with x^{-2} and x with x^{-1}, and both pairs can be written in terms of $y = x + x^{-1}$, so the quartic collapses to a quadratic in y. In (ii) there is no hint, so we look for a twisted version of the same symmetry: the coefficients 1, 1, -10, -4, 16 pair up once we allow a scaling factor, which suggests the substitution $y = x - 4/x$.

Solution

Part (i)

The constant term is $1 \neq 0$, so $x = 0$ is not a root and we may divide by x^2:

$$x^2 + 10x + 26 + \frac{10}{x} + \frac{1}{x^2} = 0, \qquad \text{that is} \qquad \left(x^2 + \frac{1}{x^2}\right) + 10\left(x + \frac{1}{x}\right) + 26 = 0.$$

With $y = x + x^{-1}$ we have $y^2 = x^2 + 2 + x^{-2}$, so $x^2 + x^{-2} = y^2 - 2$. The equation becomes

$$y^2 - 2 + 10y + 26 = 0 \iff y^2 + 10y + 24 = 0 \iff (y+4)(y+6) = 0,$$

so $y = -4$ or $y = -6$. Now we return to x, multiplying $x + x^{-1} = y$ through by x (again fine, as $x \neq 0$):

$$x + \frac{1}{x} = -4 \iff x^2 + 4x + 1 = 0 \iff x = -2 \pm \sqrt{3},$$
$$x + \frac{1}{x} = -6 \iff x^2 + 6x + 1 = 0 \iff x = -3 \pm 2\sqrt{2}.$$

These are four distinct numbers, and a quartic has at most four roots, so the solutions are

$$x = -2 \pm \sqrt{3}, \qquad x = -3 \pm 2\sqrt{2}.$$

Notice that the roots come in reciprocal pairs: $(-2+\sqrt{3})(-2-\sqrt{3}) = 1$ and $(-3+2\sqrt{2})(-3-2\sqrt{2}) = 1$. That is exactly what a palindromic polynomial must do, since replacing x by $1/x$ and multiplying by x^4 gives back the same equation.

Part (ii)

Ask yourself The coefficients 1, 1, -10, -4, 16 are not symmetric. Is there a substitution like the one in (i) that still works?

Compare the outer pairs of coefficients. The constant term is $16 = 4^2$ times the leading coefficient, and the coefficient of x is -4 times the coefficient of x^3. So the terms pair up as

$$x^4 + 16 = x^2\Big(x^2 + \frac{16}{x^2}\Big), \qquad x^3 - 4x = x^2\Big(x - \frac{4}{x}\Big).$$

This is the palindromic pattern with $1/x$ replaced by $-4/x$. Since $x = 0$ is not a root (the constant term is 16), dividing by x^2 gives

$$\Big(x^2 + \frac{16}{x^2}\Big) + \Big(x - \frac{4}{x}\Big) - 10 = 0.$$

Let $y = x - 4/x$. Then $y^2 = x^2 - 8 + 16/x^2$, so $x^2 + 16/x^2 = y^2 + 8$, and

$$y^2 + 8 + y - 10 = 0 \iff y^2 + y - 2 = 0 \iff (y-1)(y+2) = 0,$$

giving $y = 1$ or $y = -2$. Multiplying $x - 4/x = y$ by x:

$$x - \frac{4}{x} = 1 \iff x^2 - x - 4 = 0 \iff x = \frac{1 \pm \sqrt{17}}{2},$$
$$x - \frac{4}{x} = -2 \iff x^2 + 2x - 4 = 0 \iff x = -1 \pm \sqrt{5}.$$

Again these are four distinct roots, so they are all of them:

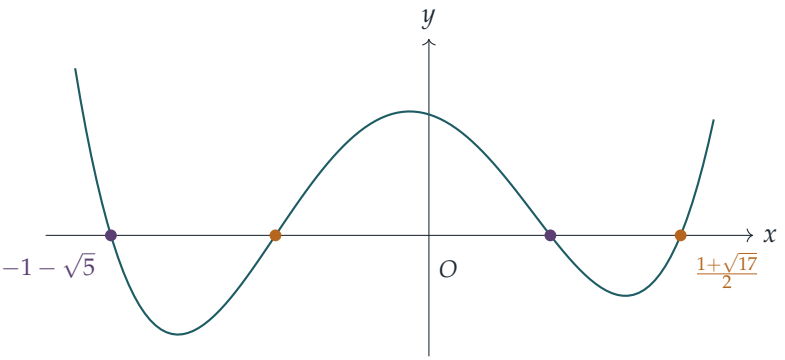
$$x = \frac{1 \pm \sqrt{17}}{2}, \qquad x = -1 \pm \sqrt{5}.$$

Here the roots pair up with product -4 rather than 1: the symmetry is now $x \mapsto -4/x$.

As a check, the working amounts to the factorisation

$$x^4 + x^3 - 10x^2 - 4x + 16 = (x^2 - x - 4)(x^2 + 2x - 4),$$

which you can confirm by expanding: the x^3 terms give $2 - 1 = 1$, the x^2 terms give $-4 - 2 - 4 = -10$, and the x terms give $4 - 8 = -4$.



The graph of the quartic in (ii) crosses the axis four times: the amber roots come from $x^2 - x - 4 = 0$ and the plum roots from $x^2 + 2x - 4 = 0$.

Checks

- Products of roots: in (i) all four roots multiply to $1 \cdot 1 = 1$, the constant term; in (ii) they multiply to $(-4)(-4) = 16$, again the constant term.
- Sums of roots: in (i) $(-4) + (-6) = -10$, minus the x^3 coefficient; in (ii) $1 + (-2) = -1$, likewise.
- Numerically, $x = -2 + \sqrt{3} \approx -0.268$ and $x = -3 - 2\sqrt{2} \approx -5.828$ both make the quartic in (i) vanish to within 10^{-9}, and $x = \frac{1}{2}(1 + \sqrt{17}) \approx 2.562$ and $x = -1 - \sqrt{5} \approx -3.236$ do the same in (ii).

Common slips

- Forgetting the cross term: $x^2 + x^{-2} = y^2 - 2$, not y^2; and in (ii) $x^2 + 16x^{-2} = y^2 + 8$, with a plus sign because the cross term of $(x - 4/x)^2$ is -8.
- Trying $y = x + 4/x$ in (ii). That gives $x^2 + 16/x^2 = y^2 - 8$ but cannot produce $x^3 - 4x$, whose signs demand $x - 4/x$.
- Stopping at the values of y. The question asks for x, and each value of y gives two values of x.

Beyond STEP

The trick works because the polynomials are invariant under a substitution: $x \mapsto 1/x$ in (i) and $x \mapsto -4/x$ in (ii). The quantity y is unchanged by that substitution, and any polynomial expression that is unchanged can be written in terms of y. This is a first glimpse of Galois theory, where the symmetries among the roots of a polynomial decide how its equation can be solved. Allenby's *Rings, Fields and Groups* is a good place to see these ideas developed.

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